# HAND GESTURE VOLUME & BRIGHTNESS CONTROL

PROJECT OVERVIEW

A Python-Based Computer Vision Application

**ACADEMIC PROJECT | B.Tech — CSBS**

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**PYTHON • OPENCV • MEDIAPIPE • PYCAW**

GitHub Repository: Hand Gesture Volume & Brightness Control

# 1. Project Overview

Hand Gesture Volume & Brightness Control is a Python-based computer vision application that allows users to interact with selected computer controls through hand gestures. A webcam captures live video, MediaPipe detects hand landmarks, and OpenCV supports image capture and processing. Python libraries are used to adjust system volume and screen brightness.

**The project demonstrates how real-time hand tracking can be connected to desktop controls to create a touch-free interaction experience.**

# 2. Problem Statement

Traditional volume and brightness controls generally require a keyboard, mouse, or direct interaction with system settings. This project explores an alternative interaction method in which supported hand gestures can trigger or adjust these controls.

# 3. Objectives

• Develop a real-time hand-tracking application using a webcam.

• Detect hand landmarks using MediaPipe.

• Use gesture information to control system volume and screen brightness.

• Provide a simple touch-free interaction method.

• Demonstrate integration of computer vision and Python system-control libraries.

# 4. Technology Stack

| Technology | Purpose |
|---|---|
| Python | Core application logic and integration |
| OpenCV | Webcam access and image/frame processing |
| MediaPipe | Hand detection and landmark tracking |
| NumPy | Numerical operations used by the application |
| Pycaw + comtypes | Windows system volume control |
| screen-brightness-control | Screen brightness adjustment |
| PyGetWindow | Window-related operations where used |

## 5. System Workflow

| | |
|---|---|
| **01** | Capture<br>The webcam captures a live video frame. |
| **02** | Detect<br>OpenCV prepares the frame and MediaPipe detects hand landmarks. |
| **03** | Interpret<br>The application reads the detected hand position or gesture. |
| **04** | Control<br>The relevant Python library updates the supported system setting. |
| **05** | Repeat<br>The process continues frame by frame while the application is running. |

## 6. Gesture Controls

The exact controls depend on the current Python script. In the version discussed during setup, an open right palm is used to maximize the window and a right fist is used to minimize it. Describe volume and brightness gestures according to the controls actually implemented in your script.

**Before presenting the project, test each gesture and update this section if the implementation has changed.**

## 7. Installation and Execution

Requirements: Windows, Python 3.10 (recommended), a webcam, and the required Python packages.

```
git clone https://github.com/yashpatil9981/Hand-Gesture-Volume-Brightness-Control.git
cd Hand-Gesture-Volume-Brightness-Control
pip install opencv-python mediapipe numpy pycaw comtypes screen-brightness-control pygetwindow
python "gesture with Brightness.py"
```

## 8. Expected Outcome

When the application runs successfully, the webcam feed is processed in real time, hand landmarks are tracked, and supported gestures are mapped to the relevant computer controls. This demonstrates touch-free interaction using computer vision.

## 9. Limitations

- A working webcam is required.
- Detection quality may vary with lighting, camera position, and hand visibility.
- Some controls and libraries are Windows-specific.
- The application must be tested with the supported gestures and installed dependencies.

## 10. Future Enhancements

- Add a calibration screen and on-screen status indicators.
- Improve gesture smoothing to reduce accidental changes.
- Add user-configurable gestures and sensitivity settings.
- Package the application for easier installation and use.

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Project Repository: Open on GitHub